

15 A student observed the effect of two different concentrations of salt solution on blood cells.

The student added each concentration of salt solution to one of two microscope slides, and then a small drop of fresh blood was added.

Each slide was viewed using the high power lens of a microscope and the student's observations were recorded.

slide 1 No red blood cells were visible.

slide 2 The red blood cells were visible but looked slightly crinkled.

Which row correctly explains the results obtained?

	slide 1	slide 2
A	Swelling of the cells caused them all to burst.	The Ψ of the cell was more negative than the Ψ of the external solution.
B	The Ψ of the cell was more negative than the Ψ of the external solution.	The Ψ of the external solution was more negative than the Ψ of the cell.
C	The Ψ of the cell was less negative than the Ψ of the external solution.	There is a net movement of water out of the cell by osmosis.
D	The Ψ of the external solution was less negative than the Ψ of the cell.	The Ψ of the cell was very similar to, but slightly more negative than, the Ψ of the external solution.

key
 Ψ = water potential